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Large-area (50 cm × 50 cm) optically transparent electromagnetic interference (EMI) shielding of ZTO/Ag/ZTO: an analytical/numerical and experimental study of optoelectrical and EMI shielding properties

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Abstract

Transparent conducting oxides (TCOs), exhibiting both high optical transparency and low electrical resistivity, are commonly employed in optoelectronic devices. However, acquiring a balance between these optical and electrical properties in a uniform way over large areas has been a pending challenge, which is essential to achieving optically transparent electromagnetic interference (EMI) shielding surfaces. In this study, we propose and demonstrate a stratified thin film structure consisting of zinc-doped tin oxide (Zn_2SnO_4, ZTO) as TCO along with a metal layer of silver (Ag) deposited on a large area of 50 cm × 50 cm polycarbonate (PC) substrate enabled by a scanning magnetron sputtering gun. We achieved high EMI shielding of 99.9% at the optical transparency of 68% in the visible spectrum by engineering the stratified architecture of ZTO/Ag/ZTO. The Ag layer of 18 nm in thickness with a sheet resistance of 10 Ω/sq yields shielding effectiveness (SE) of 27 dB in a wide frequency range of 2–20 GHz. The bottom and top ZTO layers, 20 and 40 nm thick, respectively, provide the lowest optical loss of 13% across 400–700 nm. The structure's EMI shielding, optical and structural performances were systematically characterized through a free-space focused-beam system, UV–Vis spectrophotometer, ellipsometry, focused ion-beam cross-sectional sampling and imaging, transmission electron microscopy, atomic

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force microscopy and secondary ion mass spectroscopy. EMI shielding and optical performances were validated by CST Microwave Studio and the transfer matrix method, respectively. These findings indicate that the proposed multi-layer architecture holds great promise for large-area EMI shielding and other optoelectronic applications.

Keywords: electromagnetic interference shielding, multilayer thin films, transparent conductive films, magnetron sputtering

Supplementary material for this article is available online

1. Introduction

The ability to shield electromagnetic interference (EMI) is of critical importance in electronics [1], aviation and aerospace [2–8], automotive [9–12], medical and healthcare [13–15], and telecommunications [16–18]. For example, EMI can cause signal degradation, noise, data corruption, and even system failure, jeopardizing the safety and efficiency of aircraft operations [19]. Therefore, it is essential to shield the effects of EMI and protect electronic components from malfunctioning or failure.

EMI shields are widely used particularly in aerospace applications to protect electronic systems such as radars, communications, monitoring, and avionics from EMI [7]. However, most conventional EMI shielding materials are opaque to visible light [20]. EMI shielding with high optical transmission and low electrical resistivity has been challenging, in particular over large areas. Superior transparent conducting oxide (TCO) films require high-temperature processes, which cannot be applied to heat-intolerant substrates such as polycarbonate (PC) [21]. One of the most effective approaches for achieving improved optical transmission and electrical conductivity relies on sandwiched films using TCO with noble metals (i.e. TCO/metal/TCO) [22].

The metal film's thickness is the primary factor affecting the electrical conductivity of TCO/metal/TCO structures. The electrical conductivity decreases substantially below a critical value that ranges between 5 and 15 nm [23–29] due to the discontinuities in such thin films. Besides electrical properties, significant disparities may exist between optical constants of the bulk and thin film of the same material that stems from microstructural variations [30–32]. Hence, it is crucial to consider these factors while designing and making transparent EMI shields.

Numerous studies have previously been conducted on optically transparent and electrically conductive TCO/metal/TCO structures. Some of the most commonly employed TCOs in these structures include indium tin oxide (ITO), zinc tin oxide (ZTO), zinc oxide (ZnO), Al-doped zinc oxide (AZO), and molybdenum trioxide (MoO_3), often in combination with metals such as gold (Au), silver (Ag), aluminum (Al), and copper (Cu) as the intermediate layers [33–38] However, using these materials in TCO/metal/TCO structures has limitations. The high cost of Au, the diffusion problem of Cu between layers [39], the brittleness and scarcity of indium [40, 41], the toxicity of MoO_3 [42], and the sensitivity of AZO to moisture [43] have all restricted their applicability. To address these issues, it is possible to use Ag as the metal layer and ZTO as the TCO because of their relatively low optical absorption, cost-effectiveness, and environmental friendliness [44, 45]. Several previous studies have investigated the optical and electrical properties of ZTO/Ag/ZTO multilayer films on glass and polymer substrates. However, the largest substrates utilized were limited to laboratory scales, up to 15 cm, enabled by roll-to-roll sputtering which is not applicable for rigid, large-scale, free-form objects [46–51]. Their EMI shielding in a broad frequency range at a large scale, which needs a continuous and uniform coating network over a large area, has never been reported to date. In addition, their systematic studies for the optical response both numerically and experimentally using the measured thin-film optical constants and EMI shielding properties using the measured dielectric constants completely lack in the literature.

In this work, we develop and demonstrate ZTO/Ag/ZTO (ZAZ) multilayer films deposited on a large area of 50 cm × 50 cm polycarbonate enabled by a scanning magnetron sputtering gun, unlike conventional deposition systems in which the target to be coated is fixed. This scanning sputtering gun system allows for uniform thin film coating of substrates at large scales. The thickness of the Ag layer was determined to provide EMI shielding over 99.9%, and ZTO layers were numerically optimized for the lowest optical loss.

We studied a ZTO/Ag/ZTO (ZAZ) multilayer on a PC substrate to provide EMI shielding across 2–20 GHz and high optical transmission in 400–700 nm to advance transparent EMI shielding. Moreover, the transmission spectrum between 700 and 1000 nm was investigated also for night vision. Structural, EMI shielding and optical characterizations of the ZAZ multilayer were systemically performed using focused ion-beam (FIB) cross-sectional imaging, transmission electron microscopy (TEM), atomic force microscopy (AFM), visible/near-infrared (Vis–NIR) spectrophotometry, and secondary ion mass spectroscopy (SIMS) techniques.

2. Method

ZTO/Ag/ZTO (ZAZ) multilayers were deposited onto 5.8 mm thick 50 cm × 50 cm PC samples by direct-current (DC) magnetron sputtering from the targets of ZTO and Ag at room temperature (RT). The sputtering was carried out using a ZTO target with a Zn:Sn atomic ratio of 2:1, which is thermodynam-

ically stable. For the deposition, the vacuum chamber was first evacuated to a base pressure of less than 5×10^{-6} Torr using a turbo molecular pump. The distance between the targets and substrate holder was fixed at 8 cm for the ZTO target and 10 cm for the Ag target. The sputtering power was set to 75 W and 50 W for the ZTO and Ag targets, respectively. Before each deposition process, surfaces of the ZTO and Ag targets were pre-sputtered in the argon (Ar) gas flow for 10 min to remove the native oxide layer.

Multilayer thin film coatings were performed by scanning the magnetron sputtering gun movable in three dimensions (see figure S1 in the supporting information). The *x*- and *y*-axes were used for surface scanning, while the *z*-axis was utilized to set the distance between the target and sample surface. Two sputtering guns were used to obtain large-area multilayer thin film coating. Primarily, the first scanning gun containing the ZTO target was dedicated to growing the ZTO thin film. Afterward, the second scanning gun containing the Ag target was reserved for depositing the Ag thin film. Lastly, the ZTO target scanned the sample surface to grow the last ZTO thin film to terminate the stratified structure.

The optical properties of multilayer films were investigated by optical transmittance measurements in the range of 400–1000 nm using an Agilent Cary 5000 UV–Vis–NIR Spectrophotometer. Sheet resistances of the sample were measured and mapped using a Hall Effect Measurement System (NanoMagnetics Instruments-ezHEMs) in *I*–*V* mode.

Electromagnetic characterization of the sample was conducted using a free-space focused-beam system (Compass Technology, USA) in the 2–20 GHz frequency range. The EMI shielding effectiveness in dB was obtained using

$$\mathrm{SE} = 20\log\left|\frac{E_{\mathrm{in}}}{E_{\mathrm{t}}}\right| \tag{1}$$

where E_{in} and E_{t} represent the electric field strengths of incident and transmitted electromagnetic waves, respectively [52].

The thickness of each layer was determined using FIB (Nova 600i Nanolab) etch and high-resolution transmission electron microscopy (Jeol, HRTEM). The surface roughness of ZAZ on PC was observed under non-contact mode to minimize the surface damage using an AFM (Park System XE-100). AFM data was analyzed using the Gwydion software. In addition, the surface roughness was measured according to the JIS B0601:2001 standard by laser scanning confocal microscope (Keyence VKX-100). SIMS was employed using an oxygen gun with an emission of 10 mA and 4 kV ion energy to analyze diffusion between thin film layers. Grazing incidence x-ray diffraction was conducted by PANalytical x-ray diffractometer, with Cu-Kα radiation ($\lambda = 1.5406$ Å) and at a grazing angle of 0.75° between $2\theta = 20 - 70°$ with a step size of 0.08° for determining crystal structure of ZAZ–PC.

The EMI *SE* of the structure was numerically calculated using CST Microwave Studio utilizing a unit cell module in 2–20 GHz. The dielectric constant of PC (see figure S2 in the supporting information) obtained through a free-space measurement was used in the numerical calculations. ZAZ was defined as an ohmic sheet in the model since the corresponding wavelength (in mm range) is significantly longer than the film thicknesses (in nm range). The ZAZ structure was defined as an infinitely thin sheet without thickness value in mm range and its sheet resistance value was determined by the Ag layer thickness.

The transfer matrix method (TMM) [53] was used to analytically calculate the transmission of the structure across 400–1000 nm. Optical constants of each layer of the transparent EMI shield were obtained using the VASE Ellipsometer—J.A. Woollam, USA, given in figure S3 in the supporting information.

3. Results and discussion

First, the metal layer of the structure was determined to provide EMI shielding performance in 2–20 GHz. The contribution of TCO layers was neglected in determining the EMI shielding performance due to high sheet resistance value of amorphous ZTO (see figure S4 in the supporting information). Next, the optical transmission was enhanced to a possible extent, engineering the TCO layers. Lastly, thicknesses of TCO and metal layers, surface roughness, and diffusion between the layers were investigated using FIB and HRTEM, along with AFM and SIMS, respectively.

3.1. EMI shielding performance

The complex permittivity of PC was extracted from S-parameters using a free-space measurement system depicted in figure 1(a) and imported into CST Microwave Studio. The unit cell module was utilized to numerically determine the transmission loss of the stratified structure along the frequency range of 2–20 GHz. The electromagnetic wave propagates in the *z*-direction while the structure is periodic in *x*- and *y*-directions, as shown in figure 1(b). The relation between sheet resistance (in Ω/sq), transmission ($T\%$), and shielding effectiveness (in dB) are presented in figure 1(c) for varying sheet resistance values of the resistive film on 5.8 mm thick PC.

Before determining the bottom and top ZTO layers' thicknesses, the metal thickness was determined to be 18 nm to achieve a uniform sheet resistance value of 10 Ω/sq (see figure S5 in the supporting information). The shielding effectiveness of the ZAZ–PC structure was numerically and experimentally found to be 27.43 and 27.73 dB, respectively, as shown in figure 1(d). In other words, experimental and numerical results show that over 99.9% of the incoming electromagnetic wave was shielded along the 2–20 GHz frequency range. The EMI shielding mechanism was studied for both PC and PC-ZAZ with varying thickness values of PC in terms of transmission, reflection and absorption (see figure S6 in the Supporting Information). The numerical results indicate that the primary mechanism of EMI shielding involves the reflection of EM

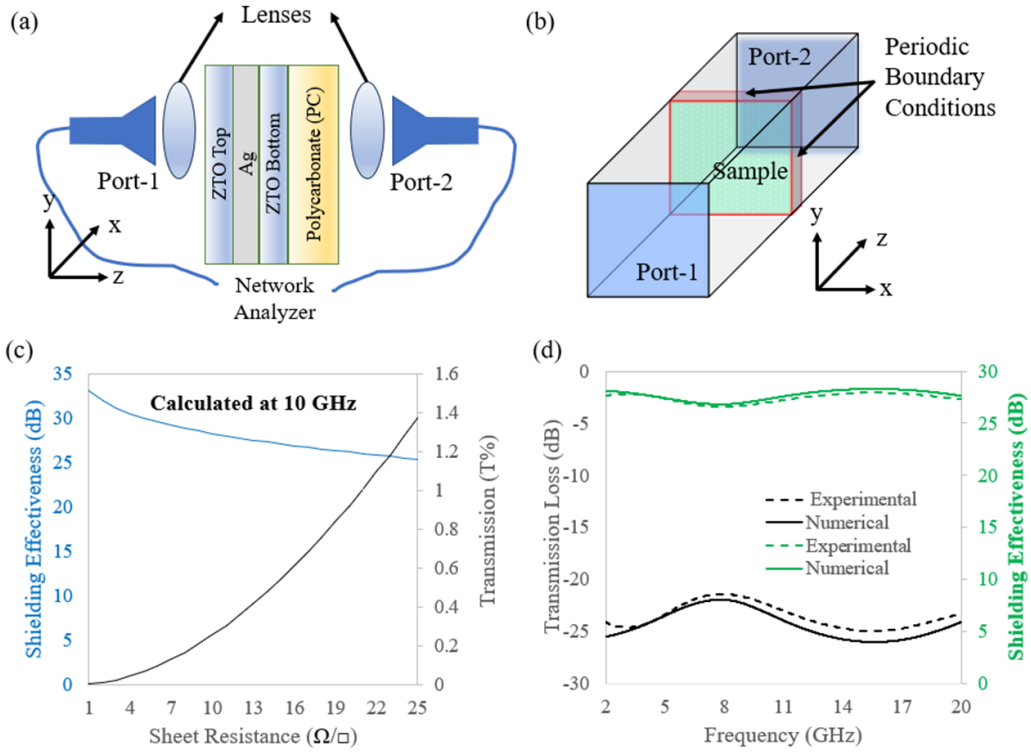


Figure 1. (a) Free-space measurement system, (b) schematic of the unit cell module, (c) sheet resistance dependent transmission loss (dB) and transmission (T%), and (d) experimentally measured and numerically computed transmission loss and shielding effectiveness of the ZAZ stratified structure on 5.8 mm thick PC over 50 cm × 50 cm area.

waves through mobile charge carriers in the metal layer incorporated with the secondary mechanism of absorption by the electric dipole [33].

3.2. Optical performance

The TMM considers multiple reflections of a multilayer structure and computes the structure's optical transmission, reflection, and absorption in multilayers, as shown in figure 2(a). The TMM was used to determine optimum thickness values of bottom and top ZTO layers for the highest transmission in 400–700 nm when the Ag film thickness was set to 18 nm to provide a sheet resistance value of 10 Ω/sq, as shown in figure 2(b). The layers were assumed infinite in the x and y directions. Calculated average transmission and reflection in the visible region of 400–700 nm, where the ZTO top and bottom layers were swept between 0 and 60 nm, are shown in figures 2(c) and (d), respectively. The optical loss due to the ZAZ coating lies below 15% in the elliptical region marked with dashed line in figure 2(c). For the highest transmission, the thickness of each ZTO layer of the ZAZ–PC structure was determined to be *ZTO bottom* (20 nm)/Ag (18 nm)/*ZTO top* 40 (nm) based on the analytical calculations, as described in the section 2.

The spectral behavior of PC with and without a ZAZ coating in 400–700 nm and 700–1000 nm ranges are shown in figures 3(a) and (b), respectively. Analytical and experimental results are consistent in the spectral ranges of both visible and near-infrared regions. Across the 400–1000 nm wavelength range, this agreement validates the accuracy of optical constants obtained through the ellipsometer, which was then used in analytical calculations. The average transmission values of the bare PC and ZAZ–PC are given in table 1 for comparison purposes. In the visible region, the analytical transmission results of PC and ZAZ–PC were found to be 82.23% and 68.34% whereas the experimental results were 82.31% and 68.80%, respectively.

On the other hand, in the near-infrared region of 700–1000 nm, the average transmission and reflection of ZAZ–PC are presented in figures 3(c) and (d), respectively. Remarkably, the ZTO top layer has more effect on optical transmission and reflection than the ZTO bottom layer in the near-infrared region. Depending on night vision requirements, the transmission and reflection of the stratified structure can be adjusted by varying the film thicknesses of ZTO layers while slightly changing the transmission in the visible region when the metal layer thickness is fixed.

3.3. Structural characterization

Figure 4(a) shows the cross-section FIB image of ZAZ–PC at room temperature. The representative image displays that the thickness of the films was uniform. The thickness of each layer was determined as ZTO top (36 nm)/Ag (18 nm)/ZTO bottom

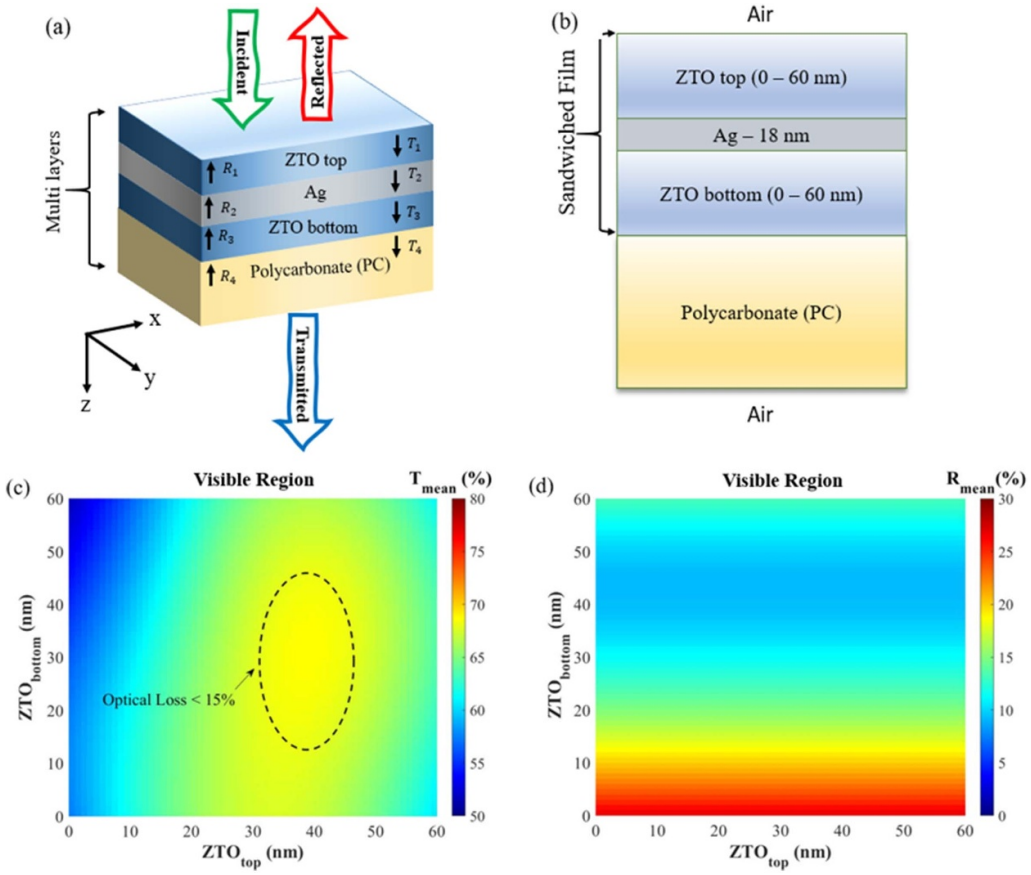


Figure 2. (a) Schematic illustration of interactions between layers; Incident (I) = Reflection (R) + Transmission (T) + Absorption (A), (b) schematic of the structure used for analytical calculations, and (c) calculated average transmission T_{mean} in 400–700 nm, and (d) calculated average reflection R_{mean} in 400–700 nm for the PC–ZTO bottom/Ag (18 nm)/ZTO top structure on PC while varying film thicknesses of the top and bottom ZTO layers.

(19 nm)–PC. The element distribution along the vertical path of the ZAZ layers was investigated using SIMS analysis, as shown in figure 4(b). Results analyzed by Gaussian functions show maxima for Zn and Sn at depths of 35 and 30 nm as well as 60 and 61 nm, respectively. The maximum of Ag occurs at 60 nm. Hence, heavier dopants penetrate a slightly shorter distance into the bottom layer, which is in good agreement with the literature [54]. The total thickness of the ZAZ structure lies around 72 nm. In addition, the thicknesses of the three layers were determined to be 20 (ZTO top), 18 (Ag) and 36 (ZTO bottom). Remarkably, these observations are in good agreement with the FIB result. However, no sharp interfaces were detected between the layers. Moreover, for 50–69 nm, a significant Ag amount was detected. Thus, a metal ion-implanted diffusion effect is observable, also previously reported [23, 55].

Figures 5(a)–(d) show 2D and 3D AFM images of the different region of the ZAZ coated surface, where root mean square (RMS) surface roughness values were 1.79, 2.23, 1.38, and 1.48 nm, respectively. AFM results indicate a uniform texture and no visible defects. RMS surface roughness values indicate a relatively smooth surface compared to the coating thickness of ZTO and Ag layers. Before the ZAZ coating, the RMS value of the PC surface was found to be 0.64 ± 0.14 nm, given in figure S7 in the supporting information. Besides, the laser scanning confocal microscopy measurement from an area of 140 μm $\times$ 1000 μm was found to exhibit an average surface roughness of 2 nm, along well with the AFM results (see figure S8 in the supporting information). These results demonstrate the potential of the optically transparent EMI shielding material to provide effective shielding while maintaining a smooth and visually appealing surface for transparent EMI applications (see figure S9 in the supporting information). X-ray diffraction (XRD) pattern of the fabricated ZTO/Ag/ZTO multilayers was given in figure S4.

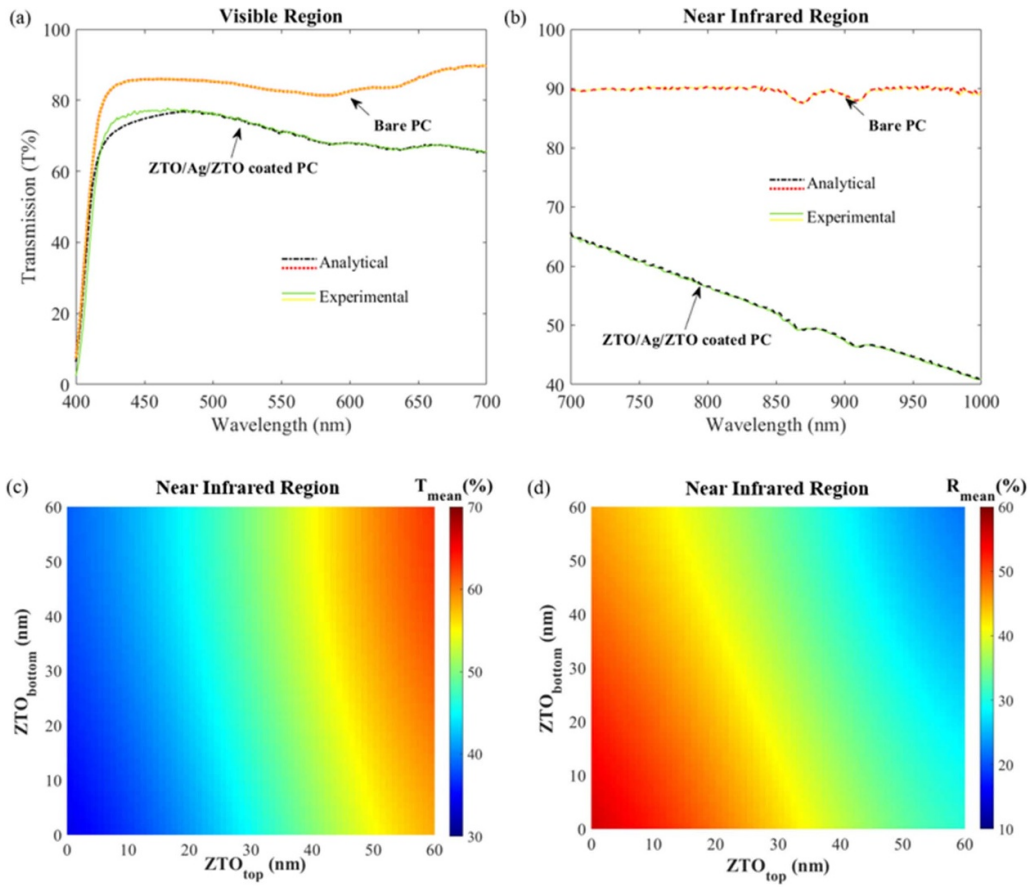


Figure 3. Spectral transmission of the ZAZ–PC structure with and without coating in the (a) visible and (b) near-infrared regions. (c) Calculated average transmission and (d) reflection of the ZTO top/Ag (18 nm)/ZTO bottom structure on PC in the near-infrared.

Table 1. Average analytical and experimental transmission (%) levels of the bare PC and ZAZ–PC in the visible (400–700 nm) and the near-infrared (700–1000 nm).

	Bare PC (%)		ZAZ–PC (%)	
	Analytical	Experimental	Analytical	Experimental
Visible region (400–700 nm)	82.23	82.31	68.34	68.80
Near infrared region (700–1000 nm)	89.65	89.67	52.49	52.49

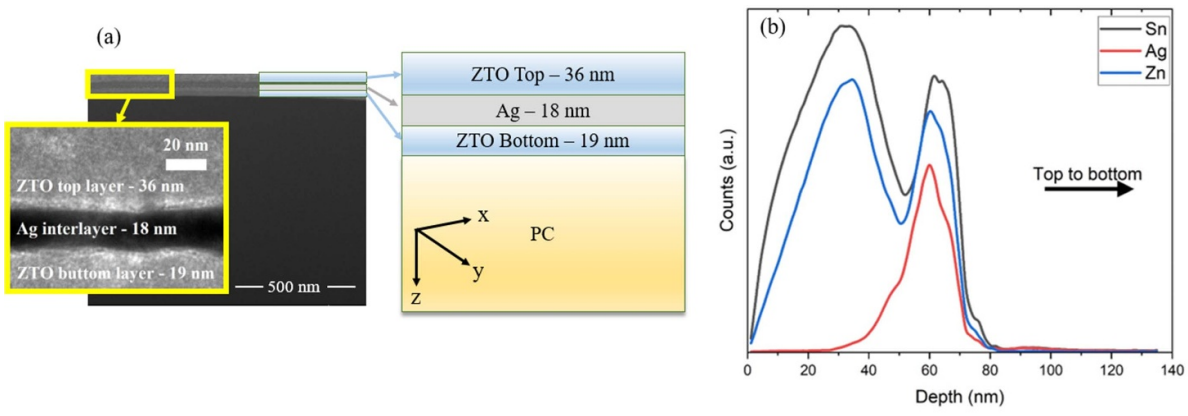


Figure 4. (a) Cross-sectional FIB image along with HRTEM inset and (b) SIMS analysis of the ZAZ–PC structure.

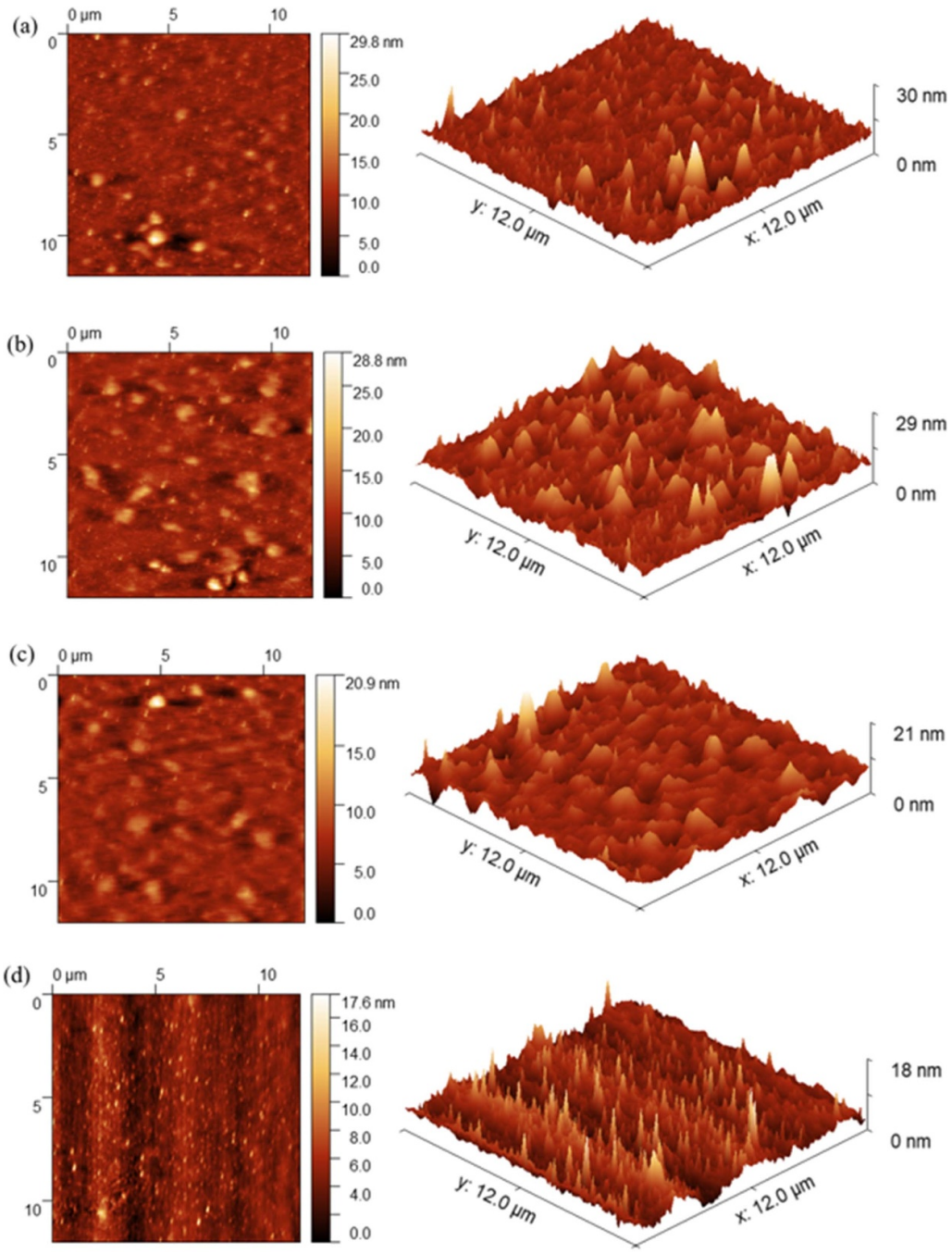


Figure 5. Representative AFM 2D and 3D topography images of different regions of the ZAZ–PC structure with RMS surface roughness values of (a) 1.79 nm, (b) 2.23 nm, (c) 1.38 nm, and (d) 1.48 nm.

4. Conclusions

This study presents EMI shielding properties of ZAZ stratified films on PC and its optical properties in 400–1000 nm over a large area of 50 cm $\times$ 50 cm. The optical and electromagnetic properties of the structure were validated thoroughly by numerical and experimental results. The ZAZ structure has a sheet resistance value of 10Ω sq^{-1}, optical loss of 13% in the visible region, and shielding effectiveness of 27 dB across the frequency range of 2–20 GHz. The optical response in the near-infrared region can be adjusted through ZTO film thickness (particularly with the top one) while keeping its EMI properties unchanged. In addition, a structural analysis of the ZAZ–PC structure was performed to examine the surface roughness and depth profile. Besides EMI shielding, ZAZ–PC structure with low RMS value, high electrical conductivity and optical transmission might be a promising candidate for other large-area optoelectronic applications.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Conflict of interest

The authors have no conflicts to disclose.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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